

## A conversational approach to medial pre-consultation

The aim of this project is to replace the free-text input field in the Catalan public health app *La meva Salut* with a structured conversational chatbot. Rather than asking patients to describe their symptoms in a blank box, the chatbot guides them step by step through a pre-consultation interview and automatically generates a report for the attending professional. The bot supports Catalan and Spanish, the two languages currently available in *La meva salut* app.

When booking a medical appointment in the health system, users are currently asked to fill in a text box explaining the reason for their visit. This unstructured approach makes it difficult for healthcare professionals to quickly extract relevant information. Our chatbot addresses this by first guiding the patient through a structured symptom collection flow. It extracts relevant clinical data such as pain location and type, intensity (from 1 to 10), duration, evolution, medication and if the patient is a smoker, has some allergies or chronic conditions, and useful extra information. Second, the bot is able to generate a pre-consultant report for the professional. Moreover, it also has the capacity to detect whether the patient is describing an emergency situation or not through the answers provided, particularly in intensity rate.

Regarding the evaluation, the system performance was tested internally across different appointments (diagnosis, sick leave, urgency, check-up and follow-up) using three metrics: flow success rate, step accuracy and language consistency. Further, different user experiences were also tested using the System Usability Scale (SUS), which reached a score of 76.1 out of 100, falling in the *Good* usability range.

During the evaluation, different limitations emerged that should be covered in future work. If the initial description is not accurate or too vague, the chatbot repeats the question until the user gives it some more context. Moreover, once the flow ends, further user input is handled inconsistently, and the flow is not resumed. In addition, some messages are sometimes duplicated, such as the medical appointment at the end of the flow. By language consistency, occasionally, there is a switch from Catalan to Spanish if the user does not approve the report generated and wants to modify some answer before submitting the pre-consulting record.

Before a real implementation of this chatbot for *La meva salut* app, there are some considerations that would need to be taken into account. Legal frameworks such as medical AI regulation should be reviewed, as the bot would handle private data, and it may be considered to extend the system to

paediatric use cases. Moreover, as the current app is connected with the security social number of each user, their medical history is already set up on each personal account and their corresponding physician for booking an appointment with their calendar if needed. Regarding languages, a future approach would also be extended to more than two, supporting English, among others.